

Exercises

Exercise 5: Exploratory analysis and graphics

Read Chapter 4 to help you complete the questions in this exercise.

1. As in previous exercises, either create a new R script or continue with your previous R script in your RStudio Project. Again, make sure you include any metadata you feel is appropriate (title, description of task, date of creation etc) and don't forget to comment out your metadata with a `#` at the beginning of the line.
2. You do not need to download anything for this exercise. You already have '*cardiacdata.txt*' in the **data** directory of your RStudio Project, from Exercise 3. If for some reason you haven't, go back to the **Data** link, download '*cardiacdata.xlsx*' and save it as a tab delimited file called '*cardiacdata.txt*' as you did before.
3. A quick reminder of what these data are. They come from a cohort study of the risk factors for cardiovascular disease, in which 163 adults aged between 55 and 75 were examined and a range of measurements taken: age and sex, systolic and diastolic blood pressure (mmHg), body mass index (kg m^{-2}), total cholesterol, HDL cholesterol and triglyceride (all mmol l^{-1}), units of alcohol drunk in the previous week, and smoking status. Note that you are importing the original file again, not the cleaned version you exported at the end of Exercise 4. Starting from the raw data every time, and doing your cleaning in a script, is what makes an analysis reproducible.
4. Import the '*cardiacdata.txt*' file into R using the `read.table()` function and assign it to a variable named `cardiac`. Use the `str()` function to display the structure of the dataset. As you found in Exercise 3, the `sex` and `smoking` variables are coded as integers, but they are really categories. Create a new variable in the `cardiac` dataframe for each of them, recoded as a factor with meaningful labels (`sex` is 1 for Female and 2 for Male; `smoking` is 1 for Current, 2 for Ex and 3 for Never). Use the `str()` function again to check the coding of these new variables. Almost every plotting function in this exercise treats a factor differently from a number, so this step is not optional housekeeping - get it wrong and your boxplots will come out as nonsense.
5. How many patients are there in each combination of smoking status and sex (hint: remember the `table()` function?)? Don't forget to use the factor recoded versions of these variables. Is the split between the smoking categories similar for women and men? Are any of the combinations small enough to worry about if you wanted to compare them? Finally, run the table again with the argument `useNA = "ifany"` so that the patients with no recorded smoking status appear as a row of their own. How many are there, and is there anything they have in common?

6. The humble cleveland dotplot is a great way of identifying if you have potential outliers in continuous variables (see Section 4.2.4). Create dotplots (using the `dotchart()` function) for the following variables; `bmi`, `systolic`, `tchol` and `alcohol`. Do these variables contain any unusually large or small observations? Don't forget, if you prefer to create a single figure with all 4 plots you can always split your plotting device into 2 rows and 2 columns (see Section 4.4 of the book).

7. That single point over on the right of the `bmi` dotplot is the value you met in Exercise 4, when you ran `summary()` and found a maximum of 514.60. A body mass index of 514 isn't possible. It's almost certainly 51.46 with the decimal point in the wrong place, although we can't confirm that. Notice how much easier the dotplot has made this to spot. In Exercise 4 you had to work your way through the `summary()` output and check the minimum and maximum of each variable in turn, whereas here it stands out as soon as you draw the plot. That's a good reason to make a dotplot of each of your continuous variables early on.

Remember that you imported the raw file again in Q4, so none of the cleaning you did in Exercise 4 is in this dataframe. Redo all three of those fixes now. Use the `which()` function to identify which observation the impossible `bmi` is, confirm the value with the square bracket `[ ]` notation, then set that `bmi`, the single `triglyceride` of 0 and the two `hdlchol` values of 0 to NA, exactly as you did in Exercise 4. Redraw the `bmi` dotchart. Now look again at all four plots from Q6. Several points still sit well away from the rest. Which ones, and what should you do about them?

8. Histograms are the standard way of looking at the distribution of a continuous variable (see Section 4.2.2). Create histograms for `bmi` and `alcohol` and put them side by side in a single figure, by splitting your plotting device into 1 row and 2 columns with `par(mfrow = c(1, 2))`. The two distributions look quite different - how would you describe them?

One thing to be careful of is that a histogram can look quite different depending on the number of 'breaks' (bins) used. The `hist()` function chooses these for you, but they're only a suggestion, so it's always worth trying a few values of the `breaks` argument before you settle on one (see `?hist`).

9. Have another look at the histogram of `alcohol` you have just drawn. A small number of patients drink a great deal more than everybody else, and those few stretch the distribution out into a long tail to the right. Data like this are described as right skewed, and they are very common in health data. It's worth doing something about, because a lot of what you might want to do with a variable later on, from summarising it to modelling it, works better when the values are more evenly spread.

One way of dealing with this is to transform the variable. `alcohol` is a count, the number of units drunk in a week, and the square root is the usual choice for count data. It also copes with the patients who drank nothing, because `sqrt(0)` is simply 0. Notice that this is a different choice from Exercise 4 Q6, where you took logs of `triglyceride`. A blood concentration and a count of units are different sorts of measurement and they don't call for the same transformation.

Create a new variable in the `cardiac` dataframe holding the square root of `alcohol`, then plot its histogram next to the untransformed one. Looking at the two, has the tail been pulled in?

10. Scatterplots are great for visualising relationships between two continuous variables (Section 4.2.1). Before you draw one, decide which variable belongs on which axis. By convention the explanatory variable goes on the x axis and the response goes on the y axis, so have a think about which is which before you draw anything.

Start with `bmi` and `systolic` blood pressure. Which is the explanatory variable, and which the response? Plot them the right way round. Now do the same for `hdlchol` and `triglyceride`. Is the choice as easy this time?

11. When visualising differences in a continuous variable between levels of a factor (categorical variable) then a boxplot is your friend (avoid using bar plots - Google ‘bar plots are evil’ for more info). Create a boxplot to visualise the differences in HDL cholesterol at each level of smoking status (don’t forget to use the recoded version of this variable you created in Q4). Include x and y axis labels in your plot. Make sure you understand the anatomy of a boxplot before moving on - please ask if you’re not sure (also see Section 4.2.3 of the book).

12. An alternative to the boxplot is the violin plot, which combines a boxplot with a kernel density plot and shows you the shape of the distribution rather than just its quartiles. You will first need to install the `vioplot` package from CRAN and make it available with `library(vioplot)`. The `vioplot()` function then works in much the same way as `boxplot()`. Draw the HDL comparison from Q11 again as a violin plot and compare the two.

Every plot so far has appeared in the RStudio plot pane, which is fine while you’re exploring but no use at all when you need the figure in a document. Getting a plot out of R takes three steps. You open a graphics device that is a file rather than a window, you draw the plot exactly as you have been doing, and then you close the device with `dev.off()`. Nothing appears on screen while you do it, and if you forget the `dev.off()` your file will be unreadable (see Section 4.5 of the book). Save your violin plot as a pdf in the `output` directory you created in Exercise 1, then find the file and open it outside RStudio to check it really worked. Exercise 6 takes this further, including how to get a figure into Word or PowerPoint at a resolution that doesn’t embarrass you.

13. To explore the relationships between several continuous variables at once it’s hard to beat a pairs plot. Create a pairs plot for the variables; `age`, `systolic`, `diastolic`, `tchol`, `hdlchol`, `triglyceride` and `bmi` (see Section 4.2.5 of the book for more details). If it looks a little cramped in RStudio then click on the ‘zoom’ button in the plot viewer to see a larger version.

These take a little practice to read, so it’s worth going slowly the first time. The variable names run down the diagonal and every other panel is a scatterplot of two of your variables. To work out which two, look at the row and the column the panel sits in: the variable named in that row is on the y axis and the variable named in that column is on the x axis. This also means that the panels above the diagonal show exactly the same pairs of variables as the panels below it, just with the axes swapped over, so you only need to look at one half. Which pair of variables looks most strongly related to you, and which variable seems to be related to almost nothing?

The `pairs()` function is very flexible and you can change what is drawn in each panel, adding things like smoothers or correlation coefficients. Have a look at Section 4.2.5 of the book and the examples in `?pairs` if you would like to try this.

End of Exercise 5